

The empirical recurrence for OEIS A250948: recovering a conjecture that is not written down

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Abstract

OEIS A250948 records “Empirical recurrence of order 88”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 748$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250948 is “Number of $(6+1) \times (n+1)$ 0..2 arrays with nondecreasing $x(i,j)+x(i,j-1)$ in the i direction and nondecreasing absolute value of $x(i,j)-x(i-1,j)$ in the j direction.”. It has offset 1 and begins

13111, 128507, 1115035, 8976385, 65910564, 447407616, 2839711239, 17032477113, ...

The entry states:

Empirical recurrence of order 88 (see link above).

As of the “Last modified” line on the live entry (Sep 03 2022 23:24:22, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2187$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 748$ of the 2187, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 748$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1496$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	23	c_{23}	−46423243298380	c_{45}	47923982363777112	c_{67}	332829870919754
c_2	−154	c_{24}	−67745345057568	c_{46}	−272531750876684250	c_{68}	72470281152986
c_3	−433	c_{25}	225291617986015	c_{47}	−11654385867040072	c_{69}	−84806016730716
c_4	9978	c_{26}	155296224784416	c_{48}	264621246250869759	c_{70}	−12047146761921
c_5	−25585	c_{27}	−874493473496031	c_{49}	−22270675031453339	c_{71}	18182453855156
c_6	−222126	c_{28}	−162644463135347	c_{50}	−223924250714353530	c_{72}	1440826147762
c_7	1303957	c_{29}	2794564344720951	c_{51}	43131360961779633	c_{73}	−3246976988827
c_8	1652738	c_{30}	−526924327159747	c_{52}	165255908933431108	c_{74}	−85660824236
c_9	−28214196	c_{31}	−7471285883968509	c_{53}	−47647606120602147	c_{75}	476265785311
c_{10}	25854473	c_{32}	3718709913903248	c_{54}	−106310790432278884	c_{76}	−8864527259
c_{11}	367421520	c_{33}	16864283009045180	c_{55}	39937121094062379	c_{77}	−56290822211
c_{12}	−882665893	c_{34}	−13132916104157134	c_{56}	59516289910054658	c_{78}	3223744325
c_{13}	−3067879256	c_{35}	−32277384404933696	c_{57}	−27278960090450188	c_{79}	5218102877
c_{14}	12919743654	c_{36}	34227298379214326	c_{58}	−28910442810670675	c_{80}	−473465844
c_{15}	14667440306	c_{37}	52372441347552744	c_{59}	15610819113380704	c_{81}	−364536604
c_{16}	−125420340375	c_{38}	−71957489760846100	c_{60}	12131124630917375	c_{82}	43702248
c_{17}	968056980	c_{39}	−71619159801123524	c_{61}	−7577290775334672	c_{83}	18005104
c_{18}	896139647225	c_{40}	126499433553249786	c_{62}	−4369052130258802	c_{84}	−2617824
c_{19}	−705091467382	c_{41}	81268986264734544	c_{63}	3135465316185106	c_{85}	−559168
c_{20}	−4908736351966	c_{42}	−189403013534004934	c_{64}	1338133546959981	c_{86}	93824
c_{21}	7204524181284	c_{43}	−73709478094921972	c_{65}	−1107069164457924	c_{87}	8192
c_{22}	20865595646667	c_{44}	244098836776599908	c_{66}	−343769490994339	c_{88}	−1536

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $88 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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